



Triglyceride-induced cardiac lipotoxicity is mitigated by *Silybum marianum*

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ABSTRACT

Background and aims: *Silybum marianum* (SM) is an herbal product with cytoprotective and antioxidant properties. We have previously demonstrated that SM ameliorates ventricular remodeling and improves cardiac performance. Here, we evaluated whether SM could exert beneficial effects against cardiac lipotoxicity in a pig model of closed-chest myocardial infarction (MI).

Methods: Study 1 investigated the effect of SM administration on lipid profile and any potential SM-related adverse effects. Animals received SM or placebo during 10 days and were afterward sacrificed. Study 2 evaluated the effectiveness of SM daily administration in reducing cardiac lipotoxicity in animals subjected to a 1.5h myocardial infarction (MI), who were subsequently reperfused for 2.5h and euthanized or kept under study for three weeks and then sacrificed.

Results: Animals administered a 10-day SM regime presented a sharp decline in plasma triglyceride levels vs. controls, with no other modifications in lipid profile. The decrease in triglyceride concentration was accompanied by a marked reduction in triglyceride intestinal absorption and glycoprotein-P expression. Three weeks post-MI the triglyceride content in the ischemic myocardium of the SM-treated animals was significantly lower than in the ischemic myocardium of placebo-controls. This effect was associated with an enhanced cardiac expression of PPAR γ and triglyceride clearance receptors. This long-term SM-administration induced a lower expression of lipid receptors in subcutaneous adipose tissue. No SM-related side-effects were registered.

Conclusion: SM administration reduces plasma triglyceride levels through attenuation of triglyceride intestinal absorption and modulates cardiac lipotoxicity in the ischemic myocardium, likely contributing to improve ventricular remodeling.

1. Introduction

Herbal products have been employed as therapeutic agents for centuries in various pathologies. *Silybum marianum* (SM), commonly known as milk thistle, is a member of the *Asteraceae* family bearing high antioxidant potential and anti-inflammatory effects [1,2]. SM seed extracts are a rich source of polyphenol compounds, primarily silymarin, which comprise its main bioactive molecule, silibinin. SM has demonstrated to be a useful resource for the treatment of gastrointestinal and hepatic disorders and mushroom poisoning by *Amanita phalloides* [3]. Additionally, it has also proven to ameliorate chemotherapy-related adverse events and to modulate metabolic syndrome conditions [4,5]. In this latter regard, SM-amelioration on metabolic disturbances is

thought to be related to silymarin's potential to enhance PPAR γ pathway mediated by isosilybin A and through induction of farnesyl X receptor (FXR) transactivity, thereby inhibiting NF- κ B signaling with the consequent reduction in the inflammatory process [6–8]. Due to its beneficial properties, SM has also shown to confer protection in the setting of ischemia/reperfusion (I/R) injury in different organs (e.g., gastric mucosa [9], liver [10], kidney [11], and brain [12]) likely through a pre-conditioning-effect [13], as well as to reduce infarct size in a rat model of open-chest myocardial infarction (MI) [14]. We have recently evidenced SM potential to exert cardioprotection and limit post-MI adverse left ventricular remodeling in a pig model of closed-chest MI. These benefits were associated with lower oxidative stress, better preservation of mitochondrial function and improved tissue survival,

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favouring overall reparative fibrosis and enhanced cardiac contractility [15,16].

Elevated triglyceride plasma levels have been linked to an increased risk of cardiovascular disease [17], and myocardial triglyceride accumulation has been associated with impaired cardiac function and cardiomyocyte death [18]. We have proven that MI *per se* elicits cardiac triglyceride deposition and lipotoxicity [19]. So far, fibrates, by primarily targeting hypertriglyceridemic disorders have demonstrated to significantly reduce major adverse cardiovascular events (MACE) without impacting all-cause and cardiovascular (CV) mortality [20]. Additionally, omega-3 fatty acids (n-3 FAs) such as eicosapentaenoic acid (EPA) in monotherapy or associated with docosahexaenoic acid (DHA) are currently recommended for patients with hypertriglyceridemia [21]. Relevantly, EPA-only therapy at 4 g/d in the randomized placebo-controlled REDUCE-IT trial displayed a significant 25% reduction in the primary end-point of a composite of CV death, nonfatal MI or stroke, coronary revascularization, or unstable angina [22]. Nonetheless, the STRENGTH trial testing a new formulation of EPA-DHA has shown no beneficial effect [23] and a previous meta-analysis involving 77,917 high-risk patients treated with n-3 FA supplementation exhibited no significant association with coronary heart disease or MACE [24]. Several mechanistic approaches have been proposed for n-3 FAs activity on triglyceride levels, in particular a reduction in hepatic triglyceride production through diacylglycerol acyl transferase as well as a decrease in triglyceride availability by ApoB inhibition and cellular lipogenesis [21]. However, their effects on triglyceride are still a matter of controversy. Finally, dietary guidance including the Mediterranean diet, the Dietary Approaches to Stop Hypertension (DASH) diet, and paleolithic diets has shown a modest impact on lipid profile and triglyceride levels [25,26].

In the present study, we aimed to evaluate the efficacy of SM on reducing triglyceride levels and preventing cardiac lipid infiltration post-MI.

2. Materials and methods

Experimental procedures were reviewed and approved by the Institutional Animal Care and Use Committees (CEEA-IR) and authorized by the Animal Experimental Committee of the local government (#5601) following the Spanish law (RD 53/2013), the European Directive 2010/63/EU and the ARRIVE guidelines [27].

2.1. Study design

The experimental protocol was conducted in healthy cross-bred commercial swine (n = 36; 4-month old Landrace x Largewhite) fed a regular (normocholesterolemic) pig chow diet. Animals were randomly allocated to the different arms and treatments and blindly received standard gelatin capsules (placebo pills) or SM (Simafit Capsules; Sanct Bernhard) before MI-induction.

The experimental design was divided into 2 studies (Supplemental Fig. 1):

- Study 1 aimed to evaluate the effect of SM administration on triglyceride absorption and lipid profile and to exclude any potential effect elicited by SM-administration on healthy hearts (i.e., without MI-induction). To that end, animals were administered SM for 10 days (Daily-SM; 15 mg/kg; 70% silibinin, equivalent to 10.5 mg/kg of silibinin extract/day; n = 6) or gelatin placebo capsules (Control; n = 6). At day 10 and 2h after SM administration, all pigs were sacrificed, and samples from the small intestine, liver, and myocardium were obtained from the anterior left ventricular wall (as done in the ischemic myocardium) and the posterior left ventricular wall (as done in the non-ischemic myocardium).
- Study 2 examined whether the administration of SM attenuates cardiac lipotoxicity in the setting of acute MI. For this purpose, pigs

(n = 24) were administered SM for 10 days (15 mg/kg; n = 12) or placebo to serve as controls (n = 12). At day 10 and 2h after SM administration, all pigs were subjected to 1.5h of complete percutaneous balloon occlusion of the mid-left anterior descending coronary artery. After the ischemic period, animals were reperfused for 2.5h and then either directly sacrificed (Control, n = 6; SM, n = 6) or allowed to recover and kept for three weeks with the same treatment regime (post-MI-SM, n = 6; post-MI-control, n = 6).

The daily dosage of SM used was comparable to those in clinical trials where SM was tested for tissue-protective effects [28].

2.2. Effect of SM on plasma lipids, liver and renal function

Blood samples were collected at baseline (prior SM administration), at day 10 (MI-induction), and sacrifice (3 weeks post-MI) for haematological (System 9000, Serono-Baker Diagnostics) and biochemical analysis (lipids, and liver and kidney parameters; Clima MC-15 RAL biotechnologies). Besides, a proof-of-concept study was carried out to monitor the effect of SM on lipid profile over time (48h period). For this purpose, pigs (n = 3) were orally administered SM at the same tested dose (15 mg/kg/day) for two consecutive days (t = 0h and t = 24h). Blood was drawn from the marginal ear from mildly sedated animals 1, 2, 4, 8, and 24h after SM ingestion for lipid panel assessment.

2.3. Analysis of SM effects on intestine and liver: lipid characterization by thin-layer chromatography (TLC)

Intestinal and liver samples were obtained at sacrifice (10 days; Study 1; Supplemental Fig. 1), immediately washed with PBS, and frozen in liquid N₂ for further analysis. Cholesteryl esters, triglyceride, and free cholesterol content were evaluated in the small intestine and liver following previously reported protocols, according to the method of Bligh and Dyer with minor modifications [29]. Briefly, one aliquot of the sample suspension was extracted with methanol/dichloromethane (2:1, vol/vol). After solvent removal under an N₂ stream, the lipid extract was re-dissolved in dichloromethane and one aliquot partitioned for TLC, which was performed on silica G-24 plates. Three different concentrations of standards (a mixture of cholesterol and cholesterol palmitate) were applied to each plate. The chromatographic developing solution was heptane/diethyl ether/acetic acid (74:21:4, vol/vol/vol). The spots corresponding to free cholesterol, cholesteryl esters, and triglycerides were quantified by densitometry against the standard curve of cholesterol and cholesterol palmitate, respectively, using a computing densitometer (Molecular Dynamics).

2.4. Myocardial sample collection in MI-induced animals

Myocardial tissues were obtained from our study in pigs as detailed in our recent paper by Vilahur et al. [15] and in Study 2 (Supplemental Fig. 1). At sacrifice (2.5h post-reperfusion or 3 weeks post-MI) animals were tranquillized, anesthetized, and sacrificed with an intravenous injection of 10 mL KCl 2 M. Hearts were rapidly excised and weighted. Myocardial tissue samples from the ischemic- and non-ischemic myocardium were obtained, frozen in liquid N₂, pulverized and homogenized in Tripure® or lysis buffer for RNA and protein isolation.

2.5. Analysis of myocardial lipids and cardiac lipid receptors

We characterized myocardial lipids by TLC within the ischemic and non-ischemic myocardium as detailed above. In addition, we assessed gene and protein expression of lipid scavenger receptors low-density lipoprotein receptor-related protein 1 (LRP1), LRP5, and very low-density lipoprotein receptor (VLDLR), as well as protein levels of lipoprotein lipase (LPL), and a cluster of differentiation 36 (CD36). Gene levels were assessed by real-time PCR-7000 Sequence Detection System

Table 1

SM effects on lipids, glucose levels, liver and renal function. Follow-up of plasma lipids and glucose (A), liver enzymes (B) and renal function (C) in animals administered *Silybum marianum* (SM) or control for 10 days. * $p < 0.05$ vs. baseline.

A. Plasma lipids and glucose levels			
Parameter	Timing	Healthy-Control	Healthy-SM
Cholesterol (mg/dL)	Baseline	84.7 ± 7.5	71.6 ± 10.1
	10 days	79.7 ± 7.5	85.6 ± 20.1
LDL-cholesterol (mg/dL)	Baseline	45.1 ± 2.0	35.9 ± 7.6
	10 days	39.7 ± 8.7	41.6 ± 6.1
HDL-cholesterol (mg/dL)	Baseline	34.6 ± 5.6	30.4 ± 5.3
	10 days	33.1 ± 2.9	30.1 ± 2.9
Total-cholesterol/HDL	Baseline	2.5 ± 0.2	2.4 ± 0.2
	10 days	2.4 ± 0.4	2.4 ± 0.2
Triglycerides (mg/dL)	Baseline	28.7 ± 2.5	25.8 ± 15.8
	10 days	34.0 ± 7.5	8.6 ± 1.5*
HDL/LDL	Baseline	0.7 ± 0.1	0.9 ± 0.1
	10 days	0.86 ± 0.1	0.75 ± 0.1
Glucose (mg/dL)	Baseline	84.7 ± 20.8	94.2 ± 21.0
	10 days	76.3 ± 23.1	89.0 ± 11.8
B. Liver enzymes			
Parameter	Timing	Healthy-Control	Healthy-SM
GOT (U/L)	Baseline	22.3 ± 9.0	35.8 ± 8.0
	10 days	25.0 ± 6.5	35.0 ± 8.4
GPT (U/L)	Baseline	28.5 ± 12.1	43.1 ± 14.7
	10 days	37.1 ± 11.1	37.1 ± 12.7
C. Renal function			
Parameter	Timing	Healthy-Control	Healthy-SM
Urea (mg/dL)	Baseline	20.2 ± 3.6	18.6 ± 4.0
	10 days	18.0 ± 2.5	17.5 ± 2.7
Creatinine (mg/dL)	Baseline	1.4 ± 0.4	1.5 ± 0.2
	10 days	1.5 ± 0.2	1.5 ± 0.2

of ABIPRISM (Applied Biosystems) by an assay-on demand. The threshold cycle (Ct) values were determined and normalized to the housekeeping gene 18S rRNA to adjust for equal amounts of RNA. Protein expression was normalized to β -actin or Ponceau staining and presented in arbitrary density units (AU). WB antibodies and RT-PCR primers are detailed in [Supplemental Table 1](#). Additionally, immunostaining against peroxisome proliferator-activated receptor- γ (PPAR γ) was performed in myocardial samples at 3 weeks post-MI in both SM-treated animals and placebo-control animals.

2.6. Molecular analyses in the liver and small intestine

In animals from study 1, we assessed transcript and protein levels of lipid receptors in the liver and small intestine. Besides, we also evaluated liver levels of glycoprotein P and hypoxia-inducible factor 1- α (HIF1 α).

Table 2

Proof of concept study on the effects of *Silybum marianum* (SM) on plasma lipid profile over time.

	Hours	Triglycerides (mg/dL)	Total -cholesterol (mg/dL)	HDL-cholesterol (mg/dL)	LDL-cholesterol (mg/dL)
SM-intake	0	33.6 ± 7.0	68.7 ± 1.6	41.1 ± 11.4	20.7 ± 11.4
	1	9.0 ± 1.7*	65.6 ± 10.9	43.3 ± 4.2	13.8 ± 5.8
	2	6.2 ± 1.0*	64.3 ± 10.0	38.67 ± 10.5	24.4 ± 18.7
	4	17.6 ± 6.0*	58.3 ± 11.2	46.2 ± 16.4	8.6 ± 6.9
	8	30.3 ± 12.8	66.3 ± 5.5	30.2 ± 4.4	30.1 ± 4.2
SM-intake	24	33 ± 2.5	33.0 ± 6.6	32.3 ± 1.5	32.0 ± 11.8
	25	17.0 ± 1.7*	60.6 ± 7.5	45.0 ± 16.7	29.6 ± 21.0
	26	7.1 ± 1.0*	60.7 ± 5.6	38.9 ± 2.0	20.3 ± 8.3
	28	20.2 ± 1.9*	64.0 ± 1.7	44.9 ± 8.5	14.1 ± 10.1
	32	26.3 ± 1.5	62.3 ± 4.9	39.7 ± 4.9	18.1 ± 10.7
	48	34.3 ± 3.2	64.3 ± 1.1	44.5 ± 4.2	13.3 ± 5.0

n = 3 animals. * $p < 0.05$ vs. SM-intake.

2.7. Molecular analyses in peripheral fat tissue

Based on the data obtained in the myocardium 3 weeks post-MI and to determine any effect of SM on peripheral tissues post-MI, we evaluated the expression of lipid-related receptors in all pigs' subcutaneous adipose tissue.

2.8. Statistical analysis

Continuous variables are expressed as mean ± standard deviation. After testing for normal distribution (Shapiro-Wilk test) repeated measures ANOVA and paired t -test as appropriate were used to analyze variables within each group whereas unpaired t -test was used for all single time-point measurements to compare between SM-supplemented and control animals. A value of $p < 0.05$ was considered significant. The statistical analyses were performed with the statistical software package Statview.

3. Results

3.1. SM reduces triglyceride plasma levels

As shown in [Table 1A](#), a 10-day administration of SM in normolipidemic animals was associated with a significant drop in plasma triglyceride levels (from a baseline value of 25.8 ± 15.8 mg/dL to 8.6 ± 1.5 mg/dL; $p < 0.05$). No changes in triglyceride levels were detected in control pigs (28.7 ± 2.5 mg/dL to 34.0 ± 7.5 mg/dL). As per the other lipid parameters, we observed no differences in total cholesterol, high-density lipoprotein (HDL)-cholesterol or low-density lipoprotein (LDL)-cholesterol after the 10-day regimen period and no modifications were detected neither in glucose levels ([Table 1A](#)), liver enzymes ([Table 1B](#)), nor renal function parameters ([Table 1C](#)). All animals showed comparable weight (37.9 ± 1.2 kg in SM vs 40.2 ± 1.5 kg in Controls).

The proof-of-concept 48 h kinetic study showed that cholesterol levels (total-, HDL- and LDL-cholesterol) remained unchanged throughout the entire testing period ([Table 2](#)). In contrast, triglyceride concentration was sharply diminished 1h after SM intake, remaining significantly lower up to 4h (50% reduction; $p < 0.05$ vs. $t = 0$) and returning to baseline values 8h post-intake. The same metabolic profile was detected after the second SM-dose at 24h ([Table 2](#)).

3.2. SM reduces triglyceride content and glycoprotein-P expression in the small intestine

Neutral lipid content in the small intestine of animals from study 1 is shown in [Fig. 1A](#). Triglycerides are the major neutral lipid component in the small intestine, being in the duodenum of control animals three-times the amount detected in the jejunum. Interestingly, there was a 55% reduction in triglyceride content in the small intestine of SM-

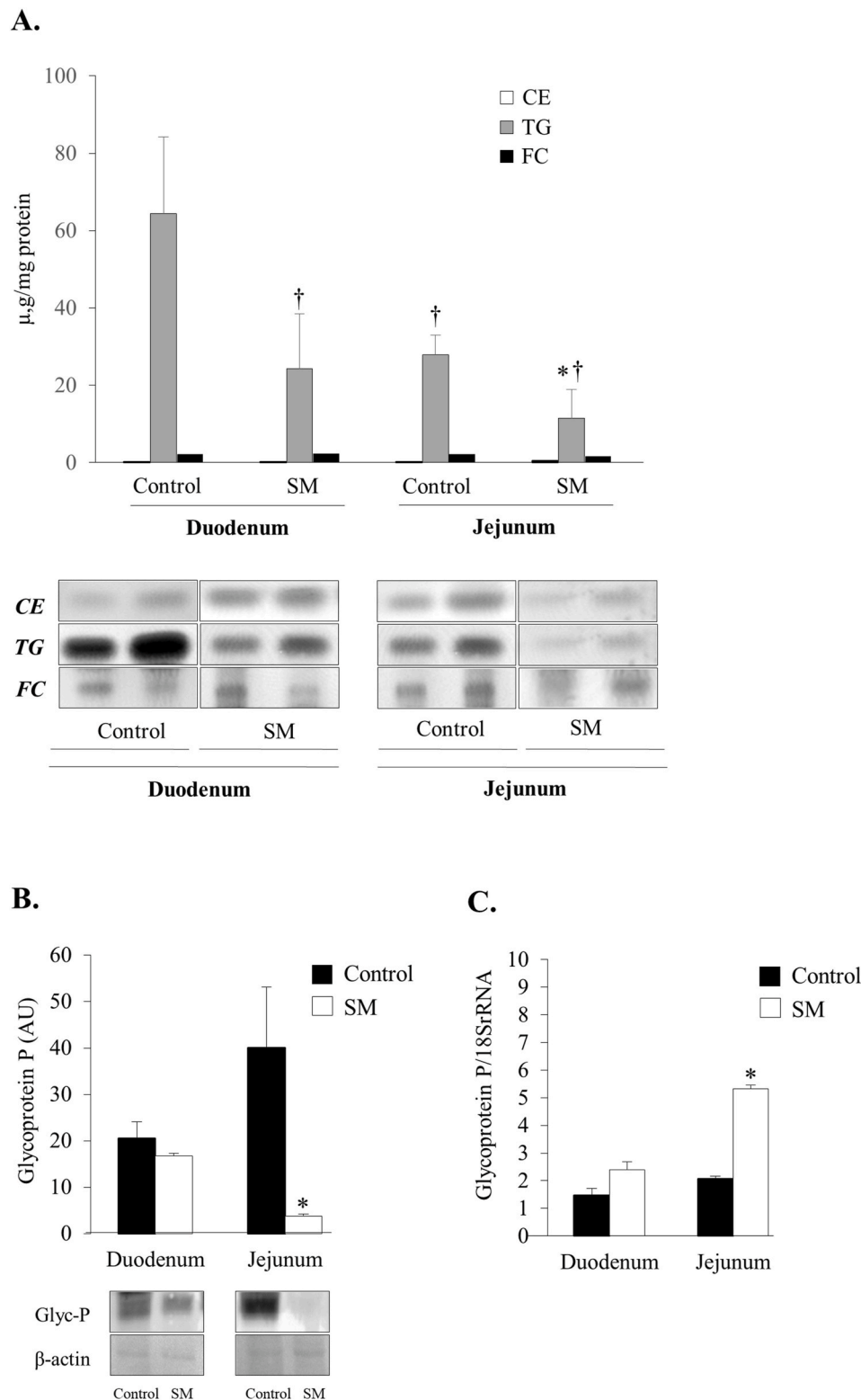


Fig. 1. Intestine lipid characterization (A) and glycoprotein-P gene (B) and protein (C) expression 10 days after SM supplementation. CE: cholesteryl esters; TG: triglycerides; FC: free cholesterol. * $p < 0.05$ vs control jejunum; † $p < 0.05$ vs control duodenum.

treated animals compared to levels in the same small intestine section of control animals (duodenum and jejunum; $p < 0.05$). Intestinal cholesteryl esters and free-cholesterol content remained unchanged among all animal groups.

Protein levels of glycoprotein-P (Fig. 1B) were found to be significantly reduced in the jejunum of SM supplemented animals, which was accompanied by the consequent induction of its transcript levels

(Fig. 1C). No changes in glycoprotein-P expression were observed in the duodenum between groups (Fig. 1B and C).

3.3. SM reduces triglyceride circulating levels without affecting lipotoxicity after ischemia/reperfusion (I/R) injury

We further investigated the impact of SM intake on circulating lipid

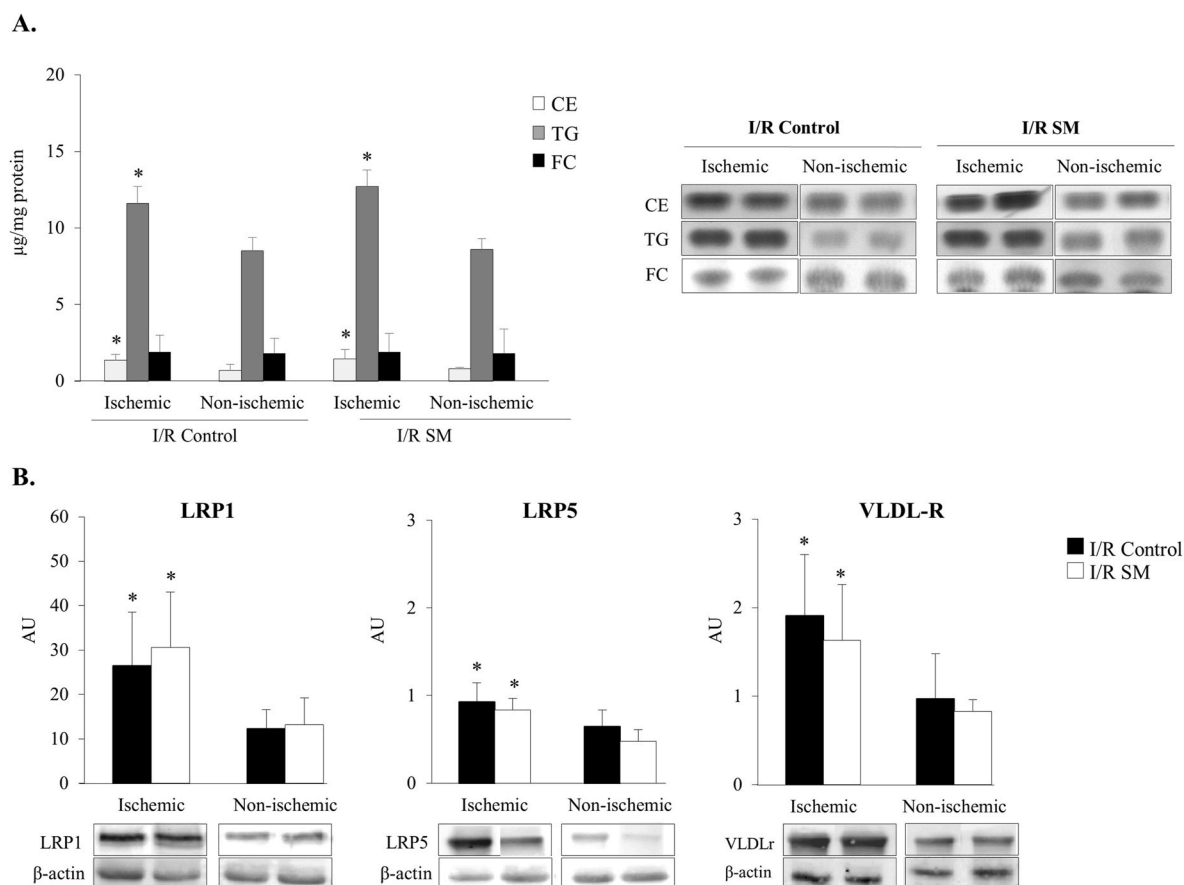


Fig. 2. *Silybum marianum* (SM) effect in cardiac lipid infiltration (A) and lipid receptors (B) early after reperfusion. (n = 6 animals/group). * $p < 0.05$ vs non-ischemic. CE: cholesteryl esters; TG: triglycerides; FC: free cholesterol.

levels in the setting of I/R. In line with our observations, administration of SM prior MI resulted in a significant acute drop in plasma triglyceride levels post SM-intake (Supplemental Table 2). We then investigated cardiac lipid content in the experimental I/R model (1.5h of complete ischemia and 2.5 h reperfusion) that resulted in an impairment of left ventricular ejection fraction (LVEF) of approximately 30%, as previously reported [20]. As shown in Fig. 2A triglyceride was the highest neutral lipid in the ischemic and non-ischemic myocardium of all animals. However, it was higher in the ischemic region as compared to non-ischemic myocardium ($p < 0.05$). Protein expression of lipid receptors was also significantly higher in the ischemic area vs non-ischemic region (Fig. 2B). During acute reperfusion, SM regimen did not result in an increased TG deposit in myocardial tissue, nor demonstrated a significant impact on lipid receptors expression (Fig. 2A and B).

3.4. Chronic SM administration after MI-induction mitigates intramyocardial triglyceride infiltration and increases cardiac expression of triglyceride clearance receptors

As shown in Supplemental Table 2, triglyceride plasma levels returned to baseline at three weeks post-MI. Of note, blood extractions were performed 12h after the last daily dose of SM. Similar to previous findings, SM-intake induced no changes in total-, HDL- or LDL-cholesterol (Supplemental Table 2), liver enzymes (Supplemental Table 3A), renal function markers (Supplemental Table 3B) and haematological parameters (Supplemental Table 4) after the 3-week administration period.

At three weeks post-MI, triglyceride content at a cardiac level was significantly augmented in the ischemia region as compared to the non-infarcted myocardium of control animals (Fig. 3; $p < 0.05$). SM

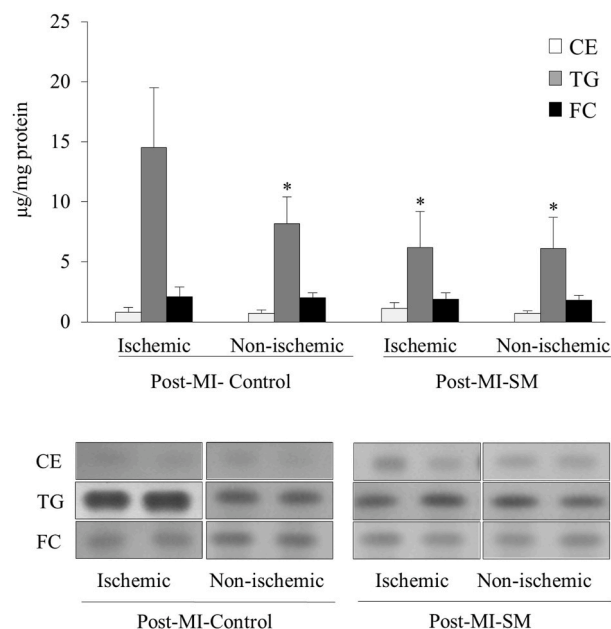


Fig. 3. *Silybum marianum* (SM) reduces cardiac lipid infiltration. MI: myocardial infarction; CE: cholesteryl esters; TG: triglyceride; FC: free cholesterol. * $p < 0.05$ vs ischemic post-MI-Control.

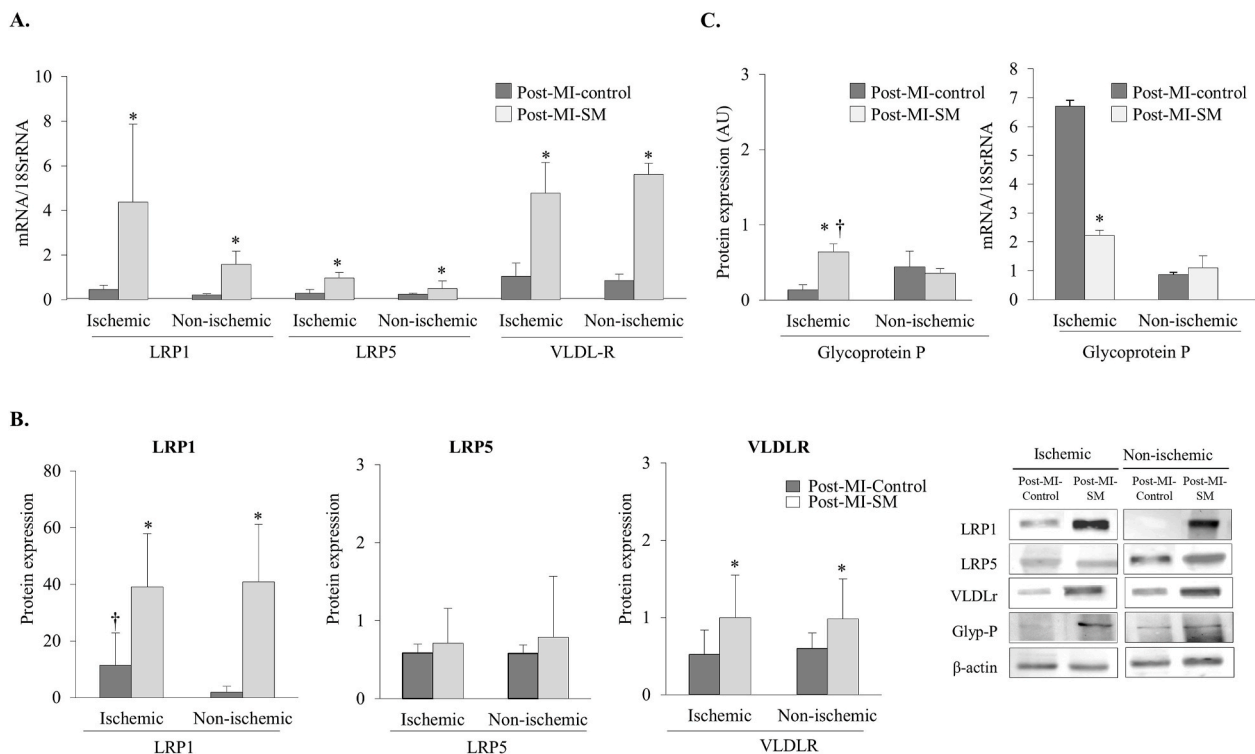


Fig. 4. *Silybum marianum* (SM) enhances cardiac lipid receptors three weeks post-myocardial infarction (MI).

Effect of SM on gene (A) and protein (B) expression of scavenger receptors LRP1, LRP5, and VLDL-R and gene and protein expression of glycoprotein-P (C) 3 weeks post-MI ($n = 6$ animals/group). MI: myocardial infarction; LRP: lipoprotein receptor-related protein; VLDL-R: very low density lipoprotein receptor. * $p < 0.05$ vs post-MI-control; $\dagger p < 0.05$ non-ischemic myocardium.

administration significantly reduced triglyceride levels in the ischemic area to values similar to the non-ischemic zone (Fig. 3).

As previously reported [20], post-MI-SM-administered animals showed a greater improvement in left ventricle ejection fraction (LVEF) as compared to post-MI-controls (post-MI vs. 3 weeks: $43 \pm 4\%$ vs. $54 \pm 3\%$ post-MI-controls and $41 \pm 2\%$ vs. $63 \pm 2\%$ post-MI-SM; $p < 0.05$). Yet, no significant correlation was detected between cardiac triglyceride content and LVEF at this time point ($r^2 = 0.13$; $y = 61.9 - 0.3x$).

Gene transcription (Fig. 4A) of lipid receptors LRP1, LRP5, and VLDL-R was significantly increased by SM in the entire left ventricle (LV). Protein levels of LRP-1, VLDL-R (Fig. 4B) were found to be higher in the entire LV of post-MI-SM animals as compared to post-MI-control animals, whereas no changes were detected for LRP5. Within the post-MI-control animals, LRP1 was enhanced in the infarcted tissue as compared to non-ischemic myocardium. Yet, expression levels were always significantly lower than those displayed in post-MI SM-treated animals.

Glycoprotein-P protein expression was significantly increased in the ischemic myocardium of SM-treated pigs compared to both control animals and the non-ischemic cardiac region (Fig. 4C). SM-treated animals had higher expression of PPAR- γ in the entire myocardial tissue (ischemic and non-ischemic regions) than placebo-treated animals (Fig. 5A). We further assessed the cardiac expression of LPL and CD36, involved in triglyceride breakdown and fatty acid (FA) uptake. As shown in Fig. 5B, protein levels of LPL were significantly reduced in the ischemic myocardium of post-MI-control compared to post-MI-SM animals. In fact, LPL levels in post-MI-SM animals were comparable to those found in non-ischemic cardiac tissue. As per CD36, SM administration for three weeks enhanced CD36 protein expression in the entire LV as compared to post-MI-controls (Fig. 5C).

3.5. SM administration does not display lipid adverse effects on healthy hearts

In animals not subjected to MI (Study 1) myocardial neutral lipids (Supplemental Fig. 2A), protein expression of CD36, LRP1, LRP5, VLDL-R, and glycoprotein-P (Supplemental Fig. 2B) were similar in the SM and control groups throughout the entire myocardium.

3.6. SM does not affect lipid content nor the expression of lipid receptors, glycoprotein-P and HIF1 α in the healthy liver

No variations were detected in the content of neutral lipids within the liver in SM-supplemented animals with respect to controls on day 10 (Supplemental Fig. 3A). Interestingly, transcript levels of LRP1, LRP5, VLDL-R, and glycoprotein-P were significantly increased in SM-treated animals (Supplemental Fig. 3B), but no differences were detected in protein levels of these receptors in unchallenged animals (Supplemental Fig. 3C). No differences were neither detected as per HIF1- α expression in both animal groups (Supplemental Fig. 3D).

3.7. SM modulates scavenger receptors protein expression in the small intestine

No changes were detected in LRP1, LRP5, VLDL-R, CD36, and FABP1 transcript levels between all animals, neither in duodenum nor jejunum (Fig. 6A). In control animals, protein levels of LRP1 and VLDL-R were found to be enhanced in the duodenum as compared to the jejunum (Fig. 6B). SM was able to significantly reduce LRP1 and VLDL-R protein expression in the duodenum, whereas it enhanced its expression in the jejunum (Fig. 6B). LRP5 and CD36 levels were unaffected by SM.

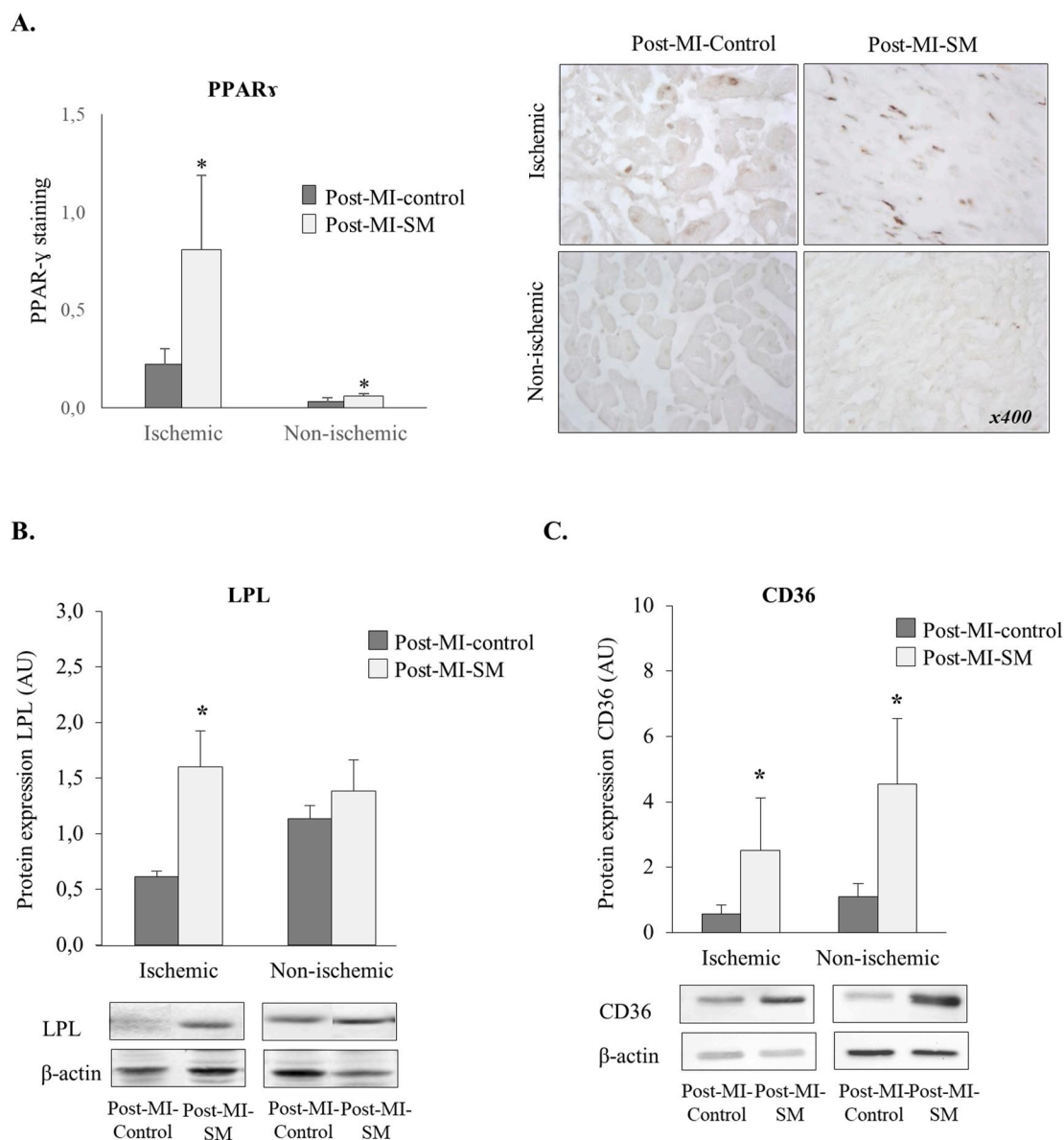


Fig. 5. Effect of SM on myocardial PPAR- γ expression (A) and LPL (B) and CD36 (C) receptors at 3 weeks post-MI. MI: myocardial infarction; PPAR- γ : peroxisome proliferator-activated receptor- γ ; LPL: lipoprotein lipase. * $p < 0.05$ vs control.

3.8. SM elicits a reduction of lipid receptors in subcutaneous adipose tissue in animals treated for three weeks post-MI

As shown in Fig. 7, SM-administered animals showed a significant decrease in VLDL-R and LPL expression in subcutaneous adipose tissue three weeks post-MI as compared to post-MI-controls. No differences were observed in LRP5 and CD36 expression within all animals over time.

4. Discussion

In the present study, we report that SM administration markedly reduces plasma triglyceride levels through attenuation of triglyceride intestinal absorption and modulates cardiac lipotoxicity in the ischemic myocardium three weeks post-infarction. So far, pre-clinical and human trials have supported SM ability to decrease triglyceride serum levels, in particular in the setting of type II diabetes and metabolic syndrome [30]. Our data suggests that SM interferes with the process of triglyceride intestinal absorption leading to a sharp decline in triglyceride plasma

levels, an effect that remains up to 4h post-ingestion. In this regard, experimental models employing specialized techniques developed to assess intestinal function, mainly Caco-2 cell permeability and single-pass intestinal perfusion, have determined that SM uptake is predominantly achieved in the small intestine [31]. Due to its high flavonoid composition, SM presents low intestinal absorption with poor bioavailability evidencing its effects mainly in the intestinal region [32, 33]. Yet, research into novel formulations to enhance its activity in targeted tissues such as liposolid compounds, bioengineered nanoparticle delivery systems and liposomal carriers is under development [34,35]. Interestingly, in subcutaneous adipose tissue we observe a significant reduction in VLDL-R and LPL protein expression at 3 weeks post-MI allowing us to discard potential confounding mechanisms of decrease triglyceride plasma levels through adipose tissue uptake. Moreover, in the small intestine of SM-treated animals we demonstrate a reduction in glycoprotein-P levels. Conflicting evidence exists as per SM effect on glycoprotein-P activity. In some clinical studies, SM has shown to hinder glycoprotein-P mediated clearance of chemotherapy drugs [36], while other findings related to the co-administration of SM with

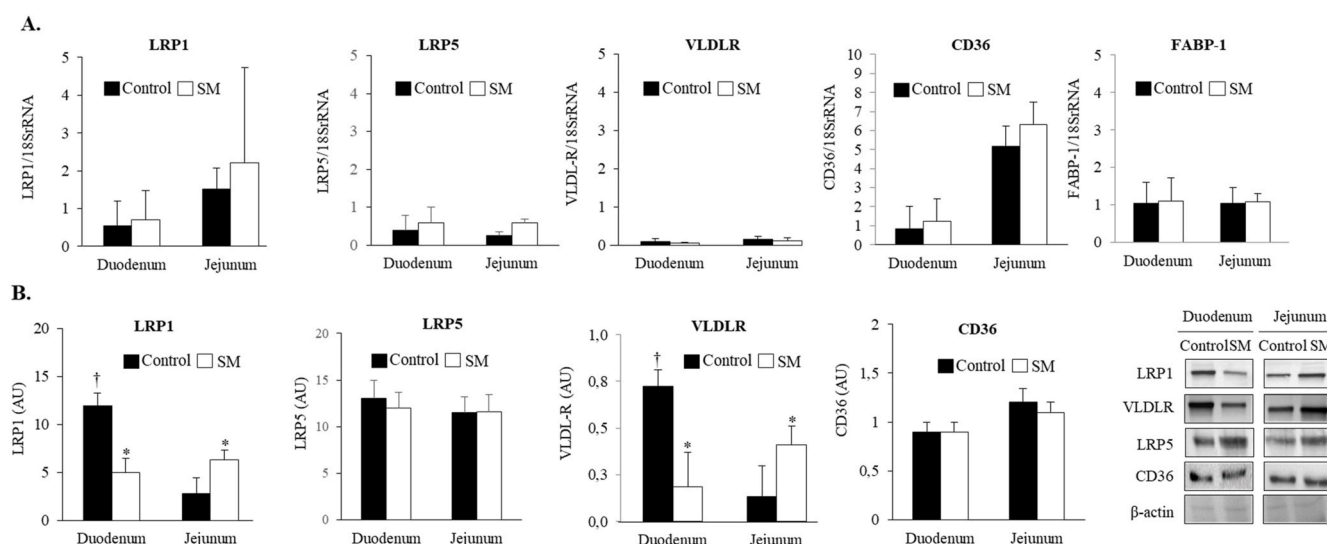


Fig. 6. Molecular analysis of lipid receptors in the small intestine.

Transcript levels (A) and protein expression (B) of lipid receptors in the small intestine of healthy animals supplemented with SM or placebo-control for 10 days. VLDL-R: very low density lipoprotein receptor; LRP: lipoprotein receptor-related protein; CD: cluster of differentiation. * $p < 0.05$ vs Control; † $p < 0.05$ vs Jejunum.

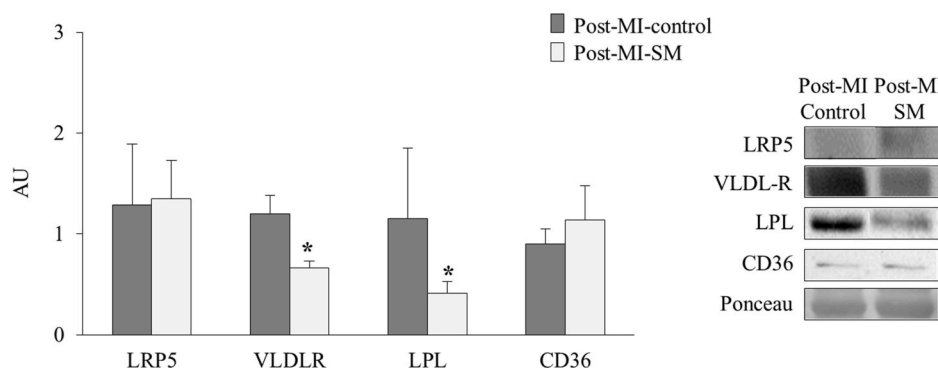


Fig. 7. Effect of SM administration on adipose tissue-related lipid receptors.

Administration of SM for 3 weeks post-MI led to a significant reduction in VLDL-R and LPL expression in the peripheral adipose tissue as compared to control animals. VLDL-R: very low density lipoprotein receptor; LPL: lipoprotein-lipase; LRP: lipoprotein receptor-related protein; CD: cluster of differentiation. * $p < 0.05$ vs Post-MI Control.

digoxin, a substrate of P-glycoprotein, displayed no variations in its pharmacokinetic profile [37]. Of note, SM has shown to impact the multidrug resistance 1 (MDR1) signaling pathway mainly through glycoprotein-P inhibition [38,39]. In fact, a study in healthy human volunteers evidenced an association between SM administration and inhibition of the MDR1 pathway [40]. Several studies, however, have pointed out the highly polymorphic condition of the MDR1 gene encoding for P-glycoprotein [41] which could result in a fluctuating *in vivo* response. Undoubtedly, further research on SM effects on MDR1 are still required.

At a cardiac level, both autopsies and imaging modalities [42] have supported the presence of excess lipid content in the infarcted area of ischemic heart disease patients, directly relating myocardial triglyceride content with ventricular dysfunction [43]. We demonstrated in a pre-clinical animal model that MI induces lipid infiltration within the jeopardized myocardium, which in turn impairs myofibroblast-induced healing of the evolving scar [19]. Besides, triglyceride accumulation in the heart has been shown to play an important role in the development of cardiac fibrosis in type II-diabetes [43]. Pre-clinical trials evaluating mitochondria-targeted antioxidants have demonstrated a crosstalk between cardiac lipotoxicity and mitochondrial oxidative stress, proving to reduce cardiac lipid damage in diet-induced obesity in rats [44]. Meanwhile, in the clinical setting, the administration of compounds known to modulate mitochondrial activity by blocking FA oxidation have decreased myocardial oxidized-FA content and insulin-resistance, ultimately enhancing cardiac function [45]. Here, we have evidence

that SM intake is not able to change cardiac triglyceride content acutely after reperfusion (2.5 h after infarction), yet, at 3-weeks after MI the daily administration of SM did result in a significant attenuation of cardiac lipotoxicity. In an earlier study [15], we detected an improvement of global ventricular performance at 3 weeks post-MI. Although we do not observe a direct correlation between cardiac triglyceride levels and cardiac performance at this time point, left ventricular remodeling remains a complex molecular and structural process extending beyond lipid deposition. Indeed, the additional benefits of SM on oxidative stress, tissue survival, and reparative fibrosis may have synergistically contributed to the general functional benefits observed on LVEF [15]. Regarding the effect on cardiac lipid receptors, these triglyceride-related structural benefits associated with SM intake were accompanied by an overall positive induction of VLDL-R and LRP1 expression within the entire LV. A study in VLDL-R and LDL-R knockout transgenic mice model has demonstrated that VLDL-R plays a key role in the peripheral uptake of triglycerides and that its deletion contributes to lipotoxicity and increased mortality [46]. Regarding SM effect on LRP1, we have earlier established that ischemia stimulates LRP1 expression in cardiomyocytes likely influencing lipid homeostasis [47]. At this time, we expand this evidence by exhibiting SM ability to further increase LRP1 activity in the injured heart. As per the role of LRP5, we do not detect changes in this receptor's expression in chronic stages post-MI, in line with our previous work [48]. However, we demonstrate that LRP5 is markedly enhanced in the ischemic myocardium in both animal groups early after reperfusion, although not being affected by SM administration. Hence, we

evidence that LRP5 seems to modulate myocardial lipid toxicity post-MI beyond its role in preventing vascular lipid infiltration [49]. Moreover, we also identify a recovery in the cardiac expression of LPL in the ischemic myocardium and higher CD36 protein expression in the entire LV of SM administered animals. LPL, through triglyceride hydrolysis, releases FAs which, upon beta-oxidation, are considered the predominant source of energy for the heart [50].

Interestingly, the hypotriglyceridemic activity of PPAR γ has been directly related to LPL induction, eventually contributing to modulate cardiac lipid metabolism [51]. In this study, we have demonstrated an enhanced expression of PPAR γ in the entire cardiac tissue of SM-treated animals in comparison to control further supporting SM-related cardiometabolic beneficial effects. At the same time, CD36 represents a key FA translocase responsible for FA cardiac uptake [52]. Of note, several studies have evaluated cardiac lipid metabolism in both LPL and CD36 knock-out mice models, displaying a significant reduction in cardiomyocyte FA uptake which is associated with an increase in plasma triglycerides levels [53,54]. Finally, we also detect an enhancement in glycoprotein-P levels in the infarcted tissue of animals under chronic SM administration. So far, an experimental study in a rat model of apnea has evidenced a significant raise in myocardial glycoprotein-P activity two weeks after intermittent exposure to hypoxia [55]. Further studies are required to determine the mechanisms by which SM may modulate glycoprotein-P expression in the cardiac tissue subjected to ischemia.

In conclusion, *Silybum marianum* acutely reduces plasma triglyceride levels as a result of decrease intestinal triglyceride absorption. Moreover, daily intake of SM for 3 weeks post-MI enhances cardiac receptors related to triglyceride metabolism and PPAR γ , ultimately leading to a reduction in myocardial triglyceride-infiltration. Because SM did not exert any effect on healthy hearts, we can postulate that SM effects on cardiac lipotoxicity in the remodeled myocardium post-MI are the result of not only lowering plasma levels but of a direct effect on triglyceride clearance in the damaged myocardium due to lipid receptors up-regulation. MI-derived tissue disruption could have passively induced the accumulation of triglycerides; however, the effects are not seen in the acute phase of MI (2.5 h reperfusion animals), when the injury is inflicted in the cardiac tissue.

It is of interest that there were no safety concerns detected during the study, as well as liver and kidney functions remained within normal range. Likewise, SM inhibition of triglyceride intestinal absorption was accompanied by differential modulation of LRP1 and VLDL-R in the different regions of the small intestine, which will require additional research to identify the significance and correlation in this finding.

Adding to our prior evidence of SM effectiveness in reducing myocardial oxidative damage, preserving mitochondrial function and improving reparative fibrosis in the heart [21], SM seems a cost-effective and promising novel therapeutic strategy not only to diminish systemic triglyceride concentrations but to prevent adverse ischemic left ventricular remodeling and the subsequent development of heart failure.

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CRediT authorship contribution statement

Gemma Vilahur: Conceptualization, Methodology, Investigation, Formal analysis, Resources, Data curation, Writing – original draft, Supervision, Project administration. **Pablo Sutelman:** Methodology,

Investigation, Formal analysis, Writing – original draft, Supervision. **Guíomar Mendieta:** Writing – original draft, Formal analysis. **Soumaya Ben-Aicha:** Investigation, Formal analysis. **María Borrell-Pages:** Investigation, Formal analysis. **Esther Peña:** Investigation, Formal analysis. **Javier Crespo:** Investigation, Formal analysis. **Laura Casaní:** Investigation, Formal analysis. **Lina Badimon:** Conceptualization, Writing Review, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atherosclerosis.2021.03.014>.

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